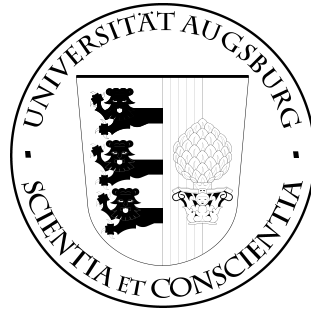


INSTITUTE OF COMPUTER SCIENCE UNIVERSITY OF AUGSBURG



Bachelor's Thesis

Title of the Bachelor's or Master's Thesis

Name

Student ID Number: Student ID Number
First Reviewer: Prof. Dr. Jakob Siegfried Kottmann
Second Reviewer: Prof. Dr. Mónica Benito
Scientific Supervisor:
Date: May 11, 2026

written at
Chair of Quantum Algorithms
Prof. Dr. Jakob Siegfried Kottmann
Institut of Computer Science
University of Augsburg
D-86135 Augsburg, Germany

Abstract

Describe the content of the present thesis here. Briefly explain the starting point of the thesis and outline its objectives. Then, discuss the methods used and the results achieved.

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1. Introduction

2. Theoretical Fundamentals of Quantum Computing

3. Excitations

General overview

3.1. Qubit Excitations

Definition and Explanation

3.2. Fermionic Excitations

Definition and Explanation

3.3. Fermionic systems

- Fermions, their challenges and their differences to qubits
- Fermionic annihilation and creation operator: $a_i^\dagger$ und a_i .

4. Different design approaches

4.1. Approach A: Operator-Algebra

- Concept: Implementation of the exact Algebra
- Structure: Addition, subtraction und multiplication of chained operators.

This approach tries to follow the exact definition of the algebra operators.

4.2. Approach B: Generator-based evolution (symbolically)

- Concept: Apply Generator directly on a state
- Unitary operation: How to calculate $U = \exp(\sum \theta_{pq} \hat{E}_{pq})$ efficient?

5. Implementation in C++

- Explanation
- Decision on implementation
- Software architecture
- Effizienz strategy:
 - Why C++ (Memory management, Speed, etc.)
 - Data structures
 - Optimization of the loop

6. Evaluation and results

- Check for Correctness with Tequila
- Performance-analyse: how does the computation scale with difference states
- Comparison of different approaches

7. Summary and Outlook

List of Figures

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Bibliography

- [1] Heise online.
- [2] C. L. Liu and J. W. Layland. Scheduling algorithms for multiprogramming in a hard-real-time environment. *Journal of the ACM*, 20(1):46–61, Jan. 1973.

A. Appendix Chapter

If you want to add information to the thesis that is only marginally relevant for understanding your work, create an appendix for this purpose. Internet reference [1], reference with multiple authors [2]

A.1. foo

A.2. bar

B. Second Appendix Chapter

